

## Teacher quick start

**Students act as Data Analysts.** They compare spectral distribution with an adequate total PPFD reading, reject “not enough total light,” diagnose selective red/deep-red rejection by the collector filter, and write CER.

|                   |                                                                                                                                                                        |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| At a glance       | Case 03 - Mars Habitat                                                                                                                                                 |
| Duration          | 60 minutes; gameplay target 18-20 minutes                                                                                                                              |
| Correct diagnosis | Light-delivery filtering removes red wavelengths needed in the game chlorophyll-biosynthesis mechanism.                                                                |
| Core evidence     | 280 PPFD; 12 m silica light pipe; 68% aggregate transmission; filter replaced 47 sols ago; 92%, 88%, 31%, and 12% band transmission; FS-7 required and BP-4 incorrect. |
| Collect           | Student Mission or selected Tasks 3, 5, 7, and 9.                                                                                                                      |
| Fallback          | Use the controlled evidence digest; keep all tasks and rubric unchanged.                                                                                               |

### 60-minute timeline

| Time  | Teacher move                                                    | Student action                                  |
|-------|-----------------------------------------------------------------|-------------------------------------------------|
| 0-5   | Launch the adequate-light/yellow-growth contradiction.          | Read mission question; predict needed evidence. |
| 5-10  | Frame quantity versus distribution without revealing diagnosis. | Complete Task 1.                                |
| 10-28 | Circulate; redirect to sensors, plants, and archive.            | Play and collect targeted evidence.             |
| 28-38 | Check exact values and no invented data.                        | Complete Tasks 2-4.                             |
| 38-50 | Prompt mechanism and alternative rejection.                     | Complete Tasks 5-7.                             |
| 50-60 | Transfer and exit ticket.                                       | Complete Tasks 8-9.                             |

**Likely sticking point:** “Adequate” describes total photon quantity, not complete spectral adequacy. Ask: Adequate total of what, distributed how?

## Formal lesson plan: outcomes and standards

### Overview

Students investigate a Mars potato-growth failure by distinguishing a scalar quantity measurement from spectral distribution. They use direct game-provided transmission values, the collector-filter record, tissue pattern, and controlled conditions to reject alternatives and explain the case mechanism.

### Measurable learning objectives

- Interpret PPFD as photon quantity over a defined waveband rather than a complete spectrum.
- Graph and compare wavelength-band transmission without inventing missing measurements.
- Use symptom location and timing to infer a failure in new pigment formation.
- Reject an alternative that conflicts with adequate total intensity and healthy roots.
- Write a concise Claim-Evidence-Reasoning explanation.

### Standards alignment

- NGSS MS-LS1-6: construct a scientific explanation based on evidence for the role of photosynthesis.
- NGSS MS-PS4-2: model reflection, absorption, and transmission of waves through materials.
- Practices: analyzing and interpreting data; constructing explanations; argument from evidence.
- Crosscutting concepts: cause and effect; systems and system models; energy and matter.

### Success criteria

- Uses  $280 \text{ } \mu\text{mol m}^{-2} \text{ s}^{-1}$  and all four exact transmission values.
- Explains that adequate total photon quantity does not guarantee adequate spectral distribution.
- Identifies red at 31% as the lowest-transmission band within 400-700 nm.
- Explains why new tissue is affected before older tissue.
- Avoids “more brightness always fixes it” and “plants use only red” claims.

## Formal lesson plan: preparation and procedure

### Materials and preparation

- Game or technical fallback digest; role-appropriate mission packet; pencil.
- Open Case 03 and confirm audio/headphone expectations.
- Print at 100% / Actual Size. Keep the Answer Key private.
- Do not preview the correct diagnosis or describe red light as the only useful wavelength.

### Procedure

| Phase           | Teacher actions                                                                                                         | Student evidence product                                  |
|-----------------|-------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| Launch          | Present the contradiction: an adequate PPFD reading with bleaching new growth. Ask what one total number cannot reveal. | Task 1: measurement limits.                               |
| Gameplay        | Prompt students to revisit crew, sensors, plants, logs, and deep archive rather than relying on symptoms alone.         | Runtime notes and exact values.                           |
| Data checkpoint | At Task 2, require only the four provided values. At Task 3, require explicit rejection of low total PPFD.              | Transmission comparison and quantity-quality explanation. |
| Diagnosis       | At Task 5, require one selected and one rejected diagnosis. At Task 6, require a complete causal chain.                 | Diagnosis and mechanism model.                            |
| CER/transfer    | Require data-specific evidence and a next measurement, not a generic "add brighter lights" fix.                         | Tasks 7-9.                                                |

### Checks for understanding

- Can the student explain why two lights with the same PPFD can have different spectra?
- Can the student identify red at 31% as the lowest transmission inside 400-700 nm?
- Does the student separate the game mechanism from broader red/blue/green plant responses?

## Teacher case analysis

### Story problem

Potato plants grew normally for about three weeks. Then top-canopy and newest leaves became pale yellow to white while older lower leaves retained green. Roots remained healthy, and iron/nitrogen additions did not help.

### Evidence channels

#### CREW

Top-down progression; nutrient additions failed.

#### SENSORS

12 m light pipes plus white LEDs; 280 PPFD.

#### PLANTS

Healthy roots; pigment failure concentrated in new tissue.

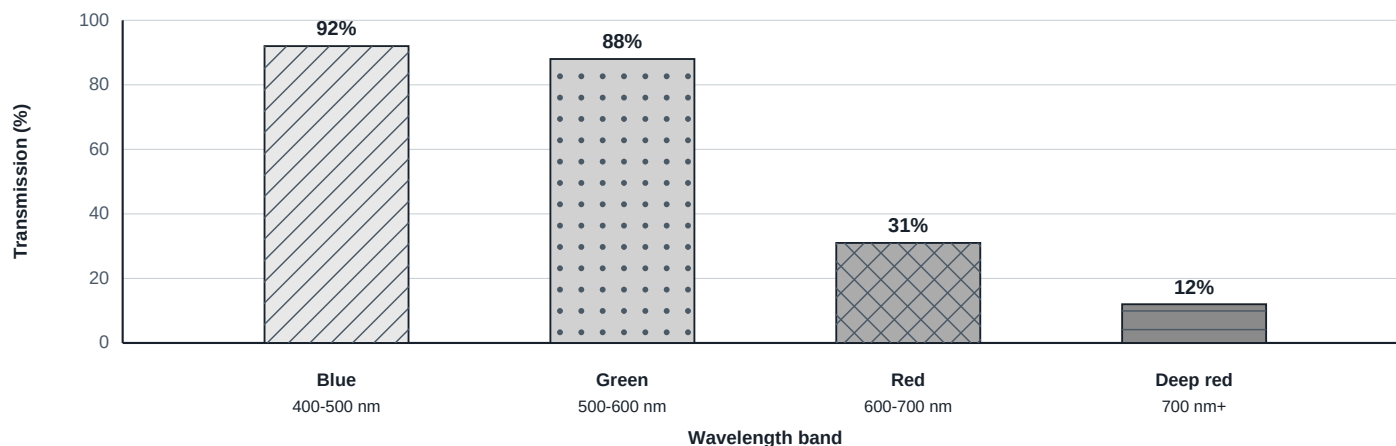
#### LOGS / ARCHIVE

Filter replaced 47 sols ago without a logged part number; FS-7 required and BP-4 incorrect.

### Essential versus supporting evidence

| Evidence                                        | Instructional weight                                |
|-------------------------------------------------|-----------------------------------------------------|
| 280 PPFD described as adequate                  | Essential: rejects simple low total light.          |
| 31% red transmission; 12% deep-red transmission | Essential: reveals selective rejection.             |
| New tissue bleaches first                       | Essential: points to new chlorophyll formation.     |
| Healthy roots; nutrients failed                 | Essential for rejecting root/nutrient alternatives. |
| 20 C, CO <sub>2</sub> 1200 ppm, UV index 0.1    | Supporting controls; do not overinterpret.          |

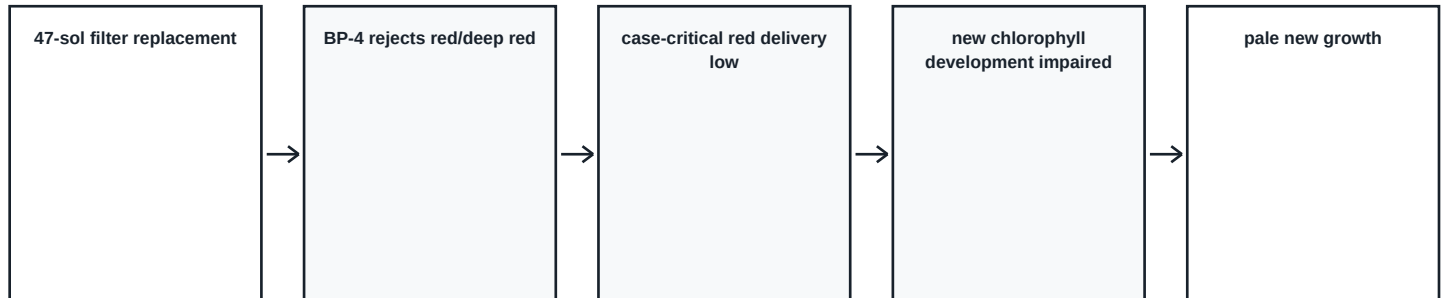
### Surface intake to habitat output transmission



**SOURCE STATUS:** Numerical bars are exact game values; general plant-light principles come from primary research.

## Mechanism, distractors, and misconceptions

### Correct case mechanism



Case mechanism model. General plant-light responses also involve blue, green, and red wavelengths.

Within the game model, the collector/light-pipe system delivers enough total photons for an adequate PPFD value while a likely BP-4 filter disproportionately rejects red/deep-red wavelengths. New tissue must form new chlorophyll; the game links the light-dependent POR step to that process. The pattern therefore appears first in new growth.

### Why the wrong answers are tempting

| Option                        | Why tempting                            | Why it fails                                                   |
|-------------------------------|-----------------------------------------|----------------------------------------------------------------|
| Perchlorates poison roots     | Mars setting activates prior knowledge. | Roots are healthy; crop uses a prepared nutrient system.       |
| CO2 too high                  | 1200 ppm sounds extreme.                | Runtime states it is within tolerance and symptoms do not fit. |
| Mars sol disrupts photoperiod | Mars day length is memorable.           | Does not explain wavelength data or new-tissue pattern.        |
| Not enough total light        | Bleaching suggests weak light.          | PPFD is 280 and described as adequate.                         |

### Misconceptions to prevent

- PPFD is a complete description of light quality.
- A brighter lamp always fixes a lighting failure.
- Green photons are useless.
- Only red light matters to every plant process.
- Yellowing automatically proves nutrient deficiency.

## Assessment, rubrics, and supports

### Quick rubric

| Dimension            | Secure                                           | Developing                          | Beginning                           |
|----------------------|--------------------------------------------------|-------------------------------------|-------------------------------------|
| Data use             | Uses 280 and all four exact transmission values. | Minor omission or error.            | Omits or invents values.            |
| Quantity vs spectrum | Clear distinction and rejection.                 | Partial distinction.                | Treats PPFD as a complete spectrum. |
| Mechanism            | Complete data-to-symptom chain.                  | Cause identified; chain incomplete. | Conclusion without mechanism.       |
| Alternatives         | Rejects with evidence.                           | Weak rejection.                     | No evaluation.                      |
| Communication        | Clear CER and vocabulary.                        | Understandable but incomplete.      | Major gaps.                         |

### Formal analytic rubric

- Accurate game data
- Graph interpretation and quantitative comparison
- Quantity-versus-distribution distinction
- Mechanistic reasoning
- Competing-diagnosis evaluation
- Source-status and uncertainty awareness
- Scientific communication and completion

### Differentiation and accessibility

- Use the six-page Accessible Mission with larger type and linear tables.
- Permit oral dictation, speech-to-text, or typed response.
- Read source-status labels aloud.
- No arithmetic is required; students compare direct measurements.
- Provide the technical fallback without reducing reasoning demand.

### Academic grading boundary

Assess evidence, mechanism, reasoning, vocabulary, alternative evaluation, and completion. Do not grade game score, speed, rapport, optional dialogue, or clue-discovery order.

## References and source-status guidance

### Authoritative and primary sources

- Illuminating Engineering Society: PPFD definition as photon quantity per area and time in the 400-700 nm band.
- McCree (1971/1972): wavelength-dependent action spectra across 22 crop species with broad blue and red maxima.
- Hogewoning et al. (2010): equal irradiance with different blue:red composition produced different plant function.
- Heyes, Ruban, and Hunter (2003): POR is a light-driven chlorophyll-biosynthesis enzyme.
- Armstrong et al. (1995): light-dependent protochlorophyllide reduction in angiosperm chlorophyll biosynthesis.
- Bula et al. (NASA/HortScience, 1991/1994): controlled-environment LED sources provide specified PAR flux and spectral regions.

### Game-specific and curriculum-original data

| Item                                                   | Status                                          |
|--------------------------------------------------------|-------------------------------------------------|
| 280 PPFD; 12 m silica pipe; aggregate transmission 68% | Game-provided Case 03 runtime data.             |
| 92%, 88%, 31%, 12% transmission                        | Direct game-provided band readings.             |
| 47-sol replacement; FS-7 and BP-4 filter models        | Game-specific maintenance and hardware records. |
| Mars symptoms and diagnosis                            | Game-specific controlled case model.            |
| Graphs and mechanism diagrams                          | Curriculum-original vector constructions.       |

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## Technical fallback and release gate

### Fallback principle

**Technical failure must not become academic failure.** Give students the same controlled evidence, exact tasks, and rubric. Do not provide the diagnosis.

### Controlled evidence digest

#### CREW

Three weeks of normal growth; newest leaves become washed out; iron and nitrogen do not help.

#### SENSORS

12 m light pipe plus white LEDs; combined PPFD 280; no spectral analysis had been performed.

#### PLANTS

Healthy roots; older leaves retain green; new growth is pale yellow-white.

#### ARCHIVE

Filter replaced 47 sols ago without a logged part number. Required FS-7; incorrect BP-4 passes blue/green and rejects most red/deep red. Transmission: 92%, 88%, 31%, 12%.

### Facilitation

- Read the digest once and allow students to revisit it.
- Require Tasks 1-9 unchanged.
- Use the same quick or formal rubric.
- Allow speech-to-text or oral response as documented accommodations; arithmetic is not required.

### Physical-print gate

Automated static, browser, PDF, portability, accessibility, overflow, checksum, and rendered-review checks may pass while the release remains unreleased. Owner physical 100%-scale printing must still verify clipping, line survival, patterns, direct labels, and writing usability before approval.

### Owner print checks

- Print each role at 100% / Actual Size on ordinary classroom equipment.
- Confirm compact page identity and role-specific N of total footers.
- Confirm graph axes, units, data labels, and patterns survive black-and-white copying.
- Confirm response boxes match expected answer length and bottom reserve remains intentional.
- Confirm no production metadata appears on ordinary printable pages.